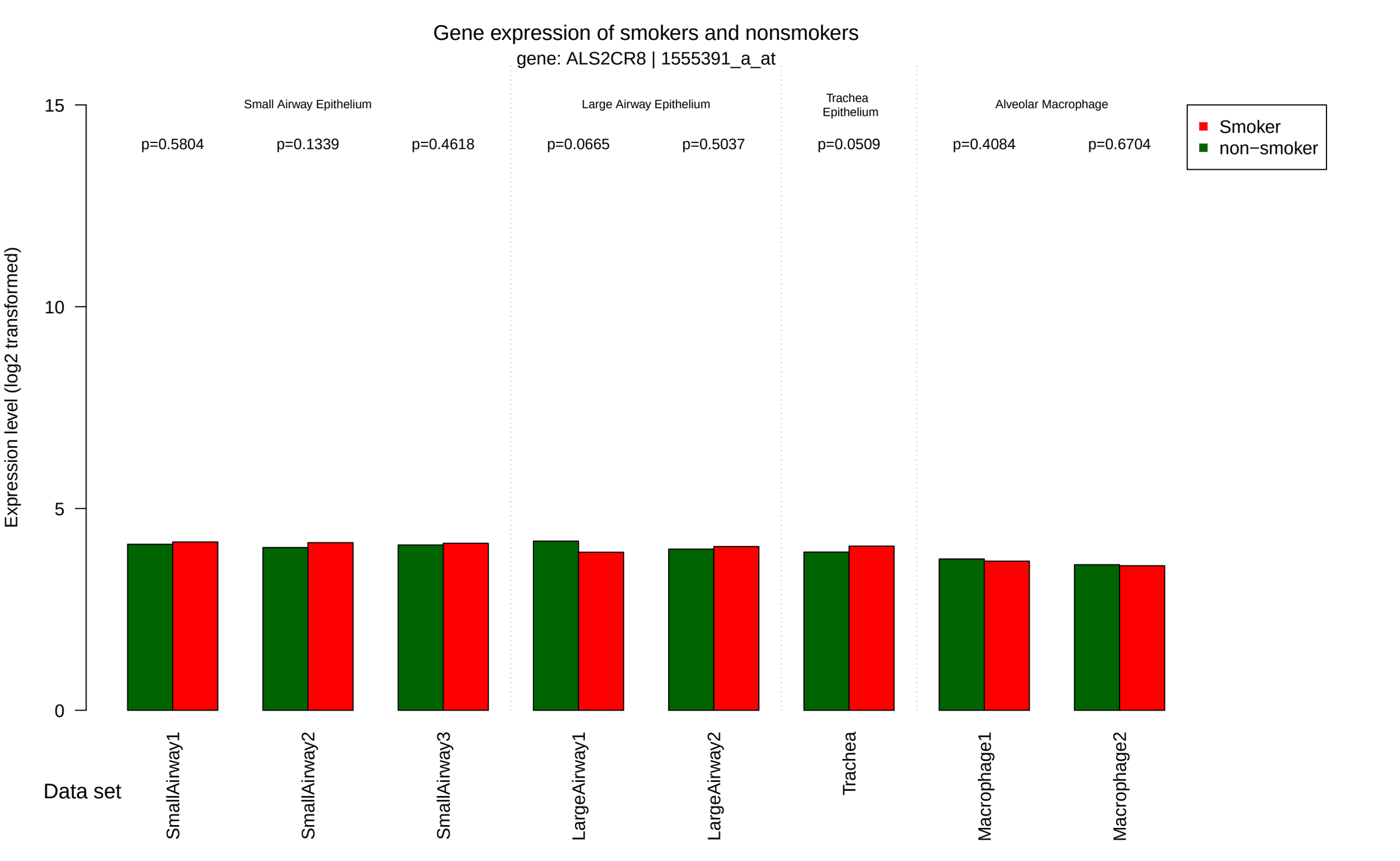
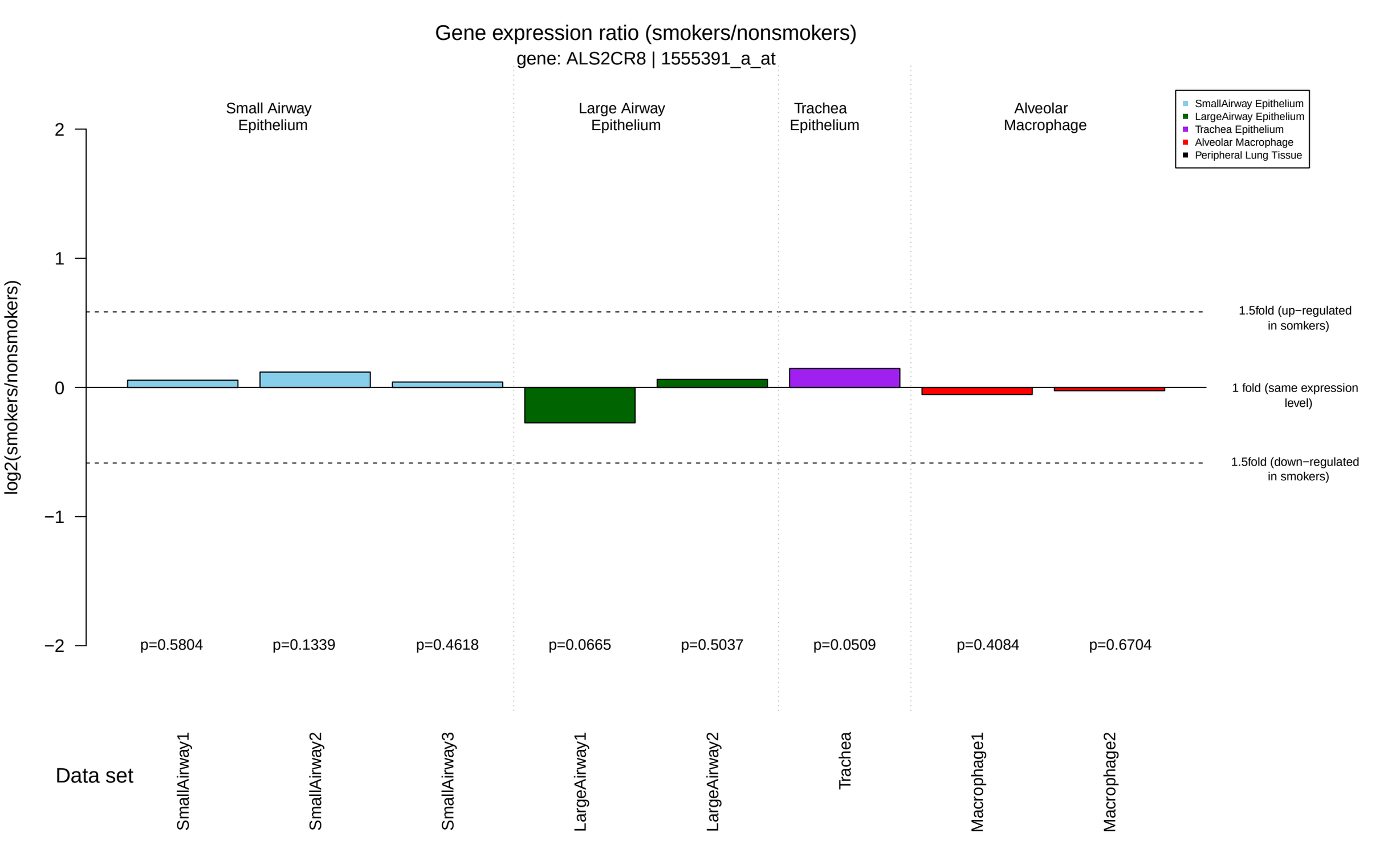
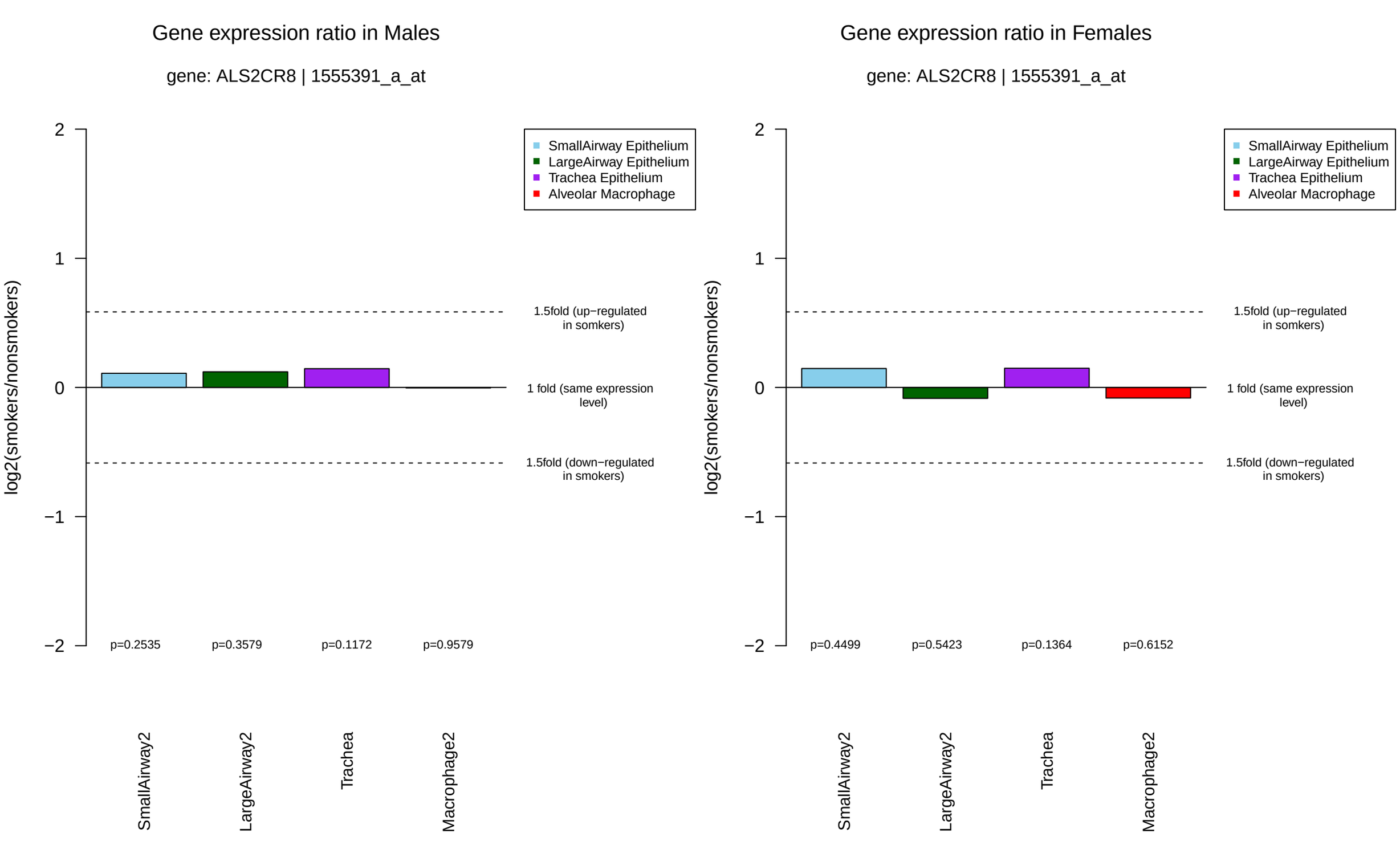
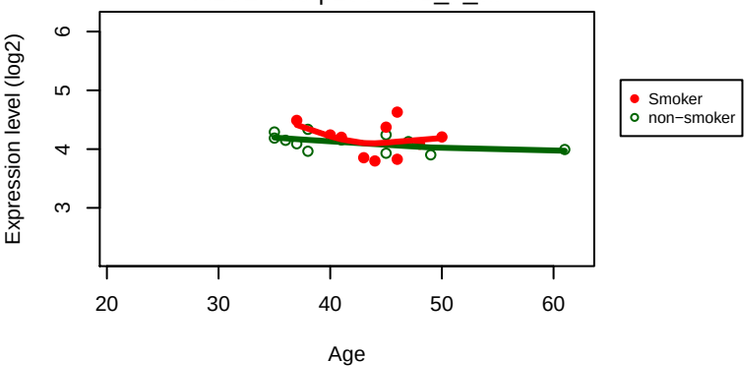
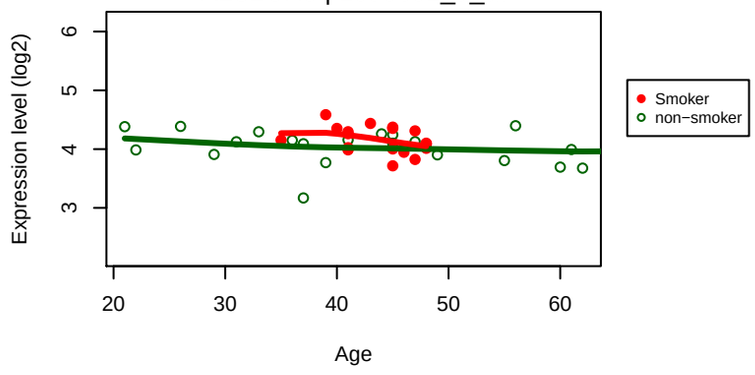
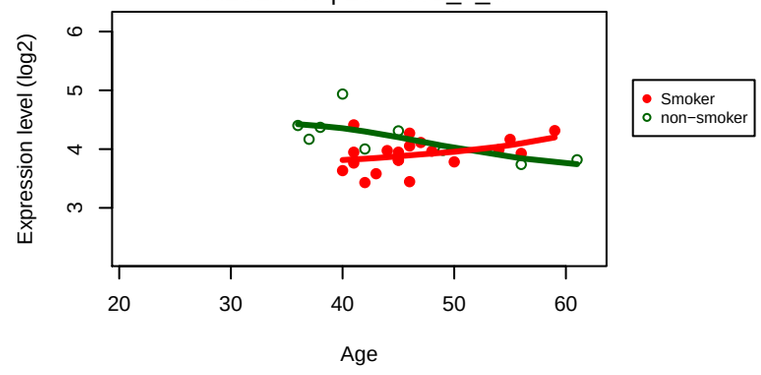

Gene expression of smokers and nonsmokers
gene: ALS2CR8 | 1555391_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5804
p=0.1339
p=0.4618
p=0.0665
p=0.5037
p=0.0509
p=0.4084
p=0.6704
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

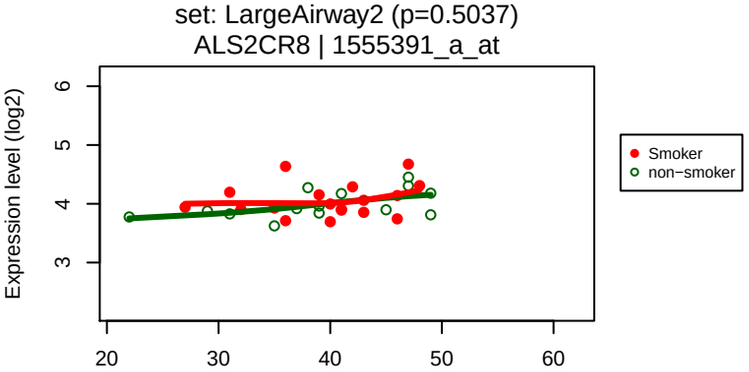

Gene expression ratio (smokers/nonsmokers)
gene: ALS2CR8 | 1555391_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5804
p=0.1339
p=0.4618
p=0.0665
p=0.5037
p=0.0509
p=0.4084
p=0.6704
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

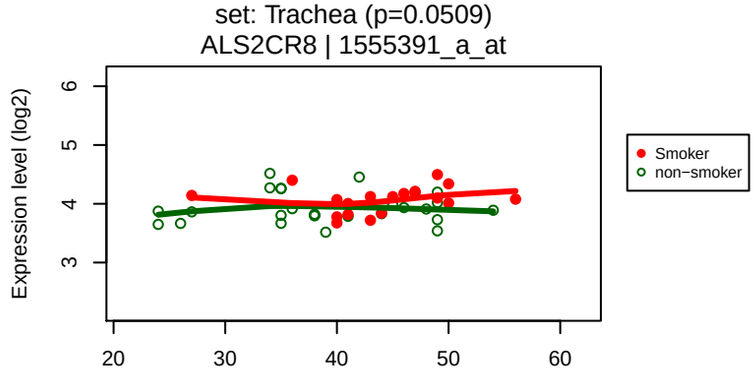

Gene expression ratio in Males
gene: ALS2CR8 | 1555391_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2535
p=0.3579
p=0.1172
p=0.9579
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALS2CR8 | 1555391_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4499
p=0.5423
p=0.1364
p=0.6152
SmallAirway2
LargeAirway2
Trachea
Macrophage2

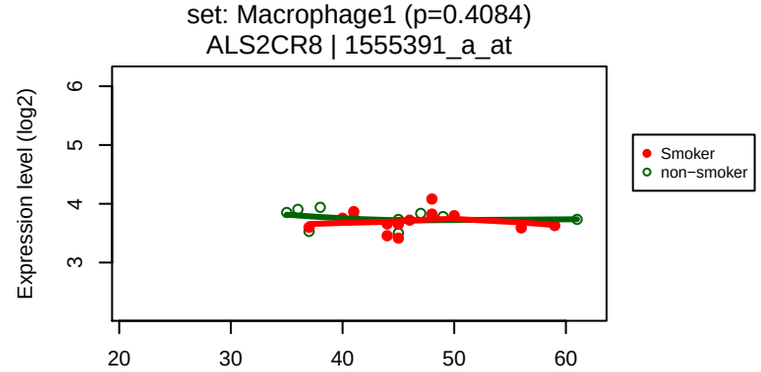

set: SmallAirway1 (p=0.5804)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker

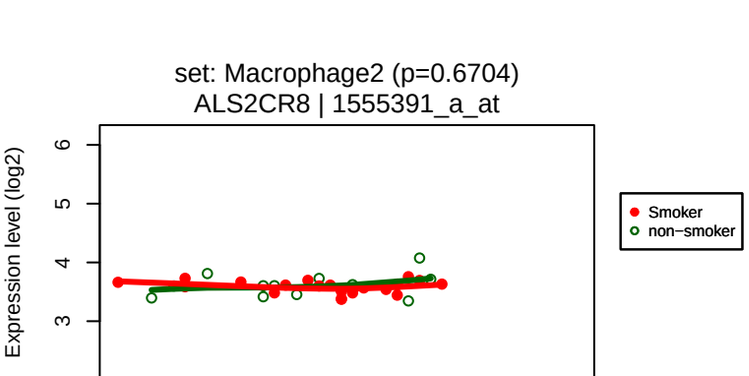

set: SmallAirway2 (p=0.1339)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker

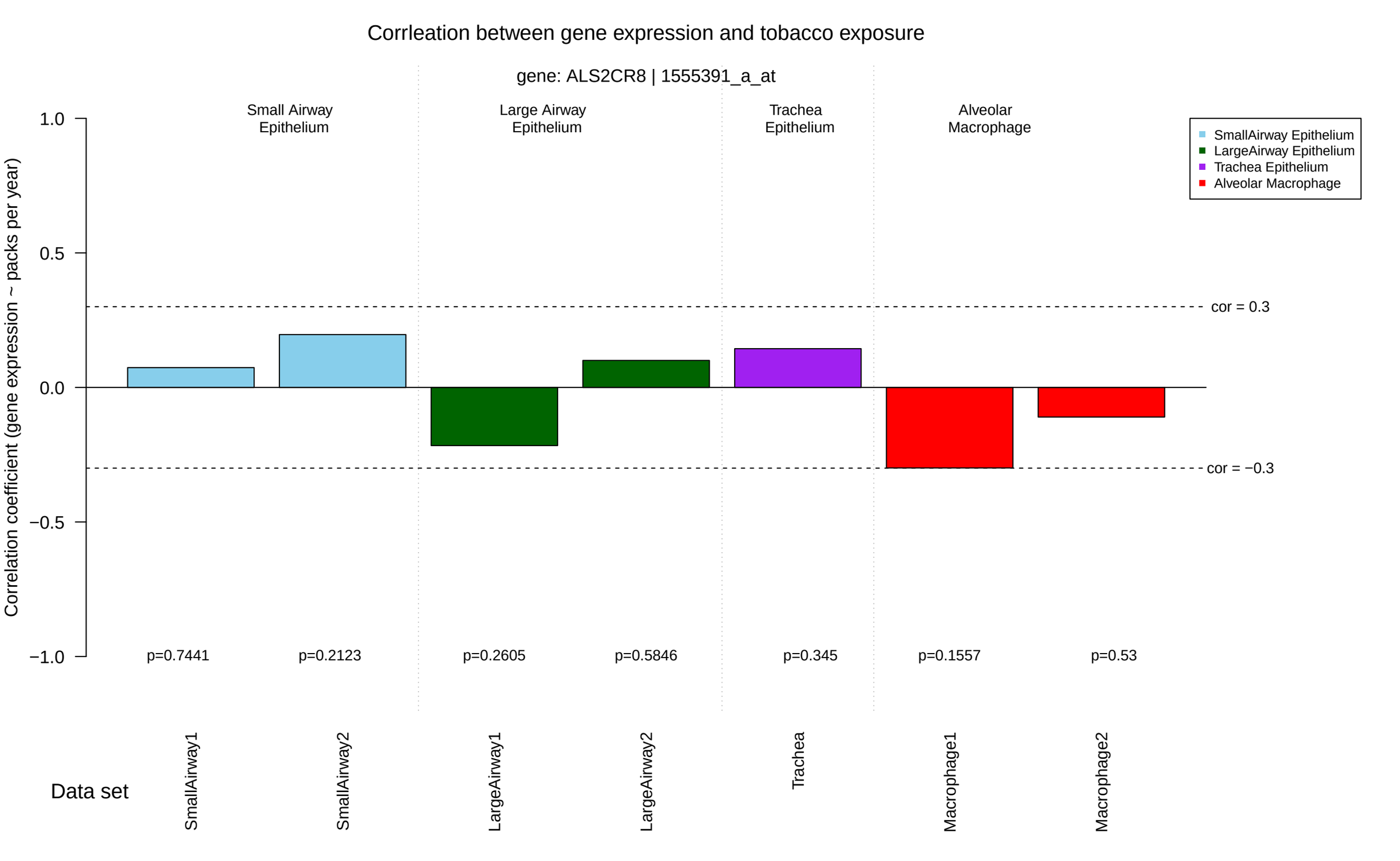

set: LargeAirway1 (p=0.0665)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5037)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0509)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4084)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6704)
ALS2CR8 | 1555391_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALS2CR8 | 1555391_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7441
p=0.2123
p=0.2605
p=0.5846
p=0.345
p=0.1557
p=0.53
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2